Low Temperature Electronic Measurements with Device Fabrication

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In this report we present a summary of the low-temperature electronic measurements undertaken during our laboratory work, which allowed us to study of the quantum Hall effect in graphene and proximity-induced superconductivity in a graphene-based superconducting quantum interference device. By looking at the resistance and conductance as a function of magnetic field and gate voltage we could plot a Landau fan and observe quantised conductance steps, which confirmed the occurance of the quantum Hall effect in our sample. Also, observing a supercurrent and investigating the oscillations in the bias current as a function of magnetic field allowed us to confirm the performance of our device, as well as estimate the area of the SQUID loop and the Josephson junctions.

I. Introduction

A. Quantum Hall Effect in Graphene

In a three dimensional system, if a charge carrying layer is exposed to a perpendicular magnetic field, B, charge accumulates at one side of the layer and depletes along the opposite. The voltage across this layer is known as the Hall voltage, discovered in 1879 by Edwin Hall [1]. The Hall resistance, the resistance corresponding to the Hall voltage, is given by:

$$R_H = -\frac{B}{qn_q d}. \tag{1}$$

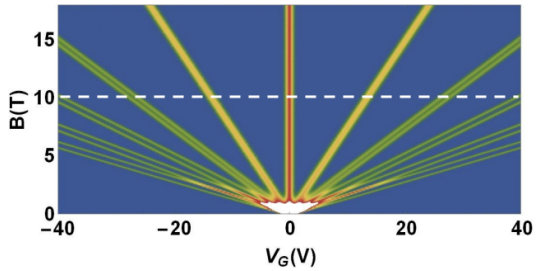
The Hall resistance is highly dependent on the properties of the conductor under examination, as it varies according to physical thickness of the sample d as well as the density of free charges n_q in Equation (1). However, in two dimensional systems, it is possible to observe a quantised version of this effect [2].

The Quantum Hall Effect (QHE) was discovered in 1985 by Klaus von Klitzing [3], a discovery for which he won the Nobel prize. The Hall resistance R_H typically varies linearly as a result of applied magnetic field [4], but at low temperatures and high magnetic fields it becomes quantised. This behaviour can be modelled as:

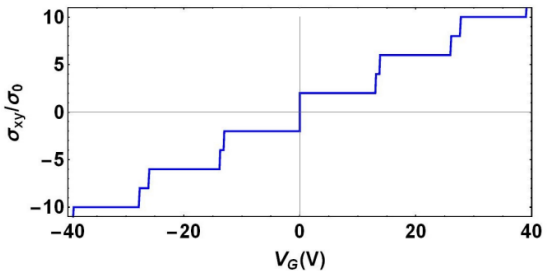
$$R_{QH} = \frac{h}{Ne^2}. \tag{2}$$

Here, h is the Planck constant, e is electron charge and N, sometimes labelled as ν, is an integer [5]. This quantisation of resistance is a result of Landau confinement within the 2D system where certain energy levels are degenerate, allowing charge carriers to exist within the same state, while others are left empty. The filled states are referred to as "Landau Levels" [2, 6]. This phenomenon can be visualised as a "Landau fan", a plot where there are clearly defined edges representing the jump from one degenerate energy level to another one, passing through all empty levels [7]. An example of such a fan can be seen in FIG. 1.

Measurements of the Hall Effect, both classical and quantum, are taken using a four-terminal set-up, shown



(a) Landau fan



(b) Conductance steps

FIG. 1: (a) Magnetic field B as a function of the gate voltage V_G. (b) Steps in the Hall conductivity at $B = 10$ T. Plots are taken from Cysne *et al.* [7].

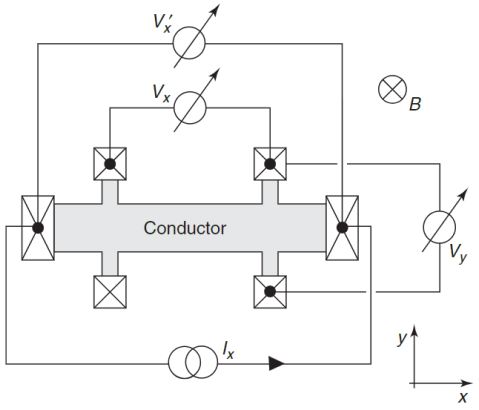


FIG. 2: Example of 4-terminal measurement setting up. Image taken from Weis *et al.* [8].

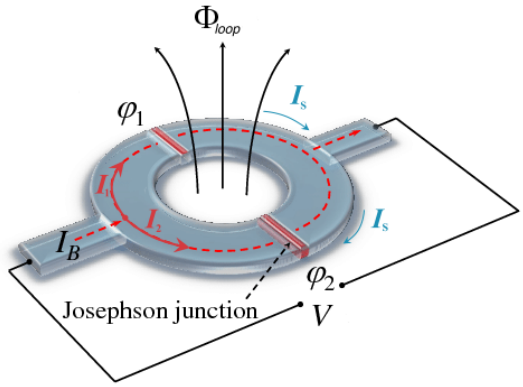


FIG. 3: Superconducting quantum interference device, with a bias current I_B applied across it which spits into I_1 and I_2 (shown in red). The screening current I_S (shown in blue) is trying to oppose the applied magnetic flux through the loop Φ_{loop}, such that the measured voltage is V. Image was taken from Bosisio *et al.* [12].

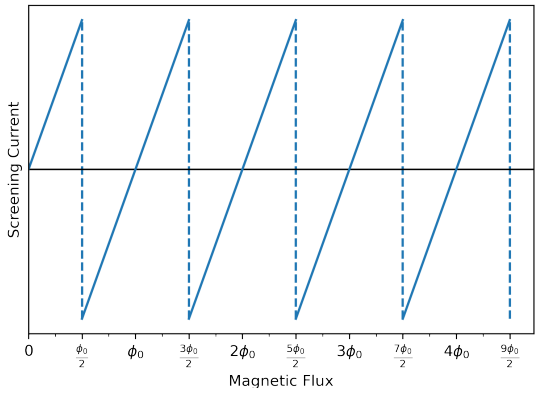
in FIG. 2. Voltage readings taken by V_x' are longitudinal measurements of the potential difference while V_y provides a transverse voltage method from which the Hall voltage can be found [8].

Graphene possesses a unique electronic structure that allows for the observation of interesting phenomena at low temperatures. Graphene's band structure takes the form of two "Dirac Cones" rather than the typical parabolic shape one would expect in similar materials. The point at which these cones connect is referred to as the "Dirac Point" [9]. Particles at this point move ultra-relativistically, but at the Fermi velocity, denoted as ν_F, instead of the speed of light. Additionally, electrons and holes are degenerate at this point; they are able to exist in the same energy levels [10]. Examination of the QHE in graphene has provided interesting insights into the behaviour of electrons at these levels and has even helped further refine SI units through showing the universality of the Ohm [11].

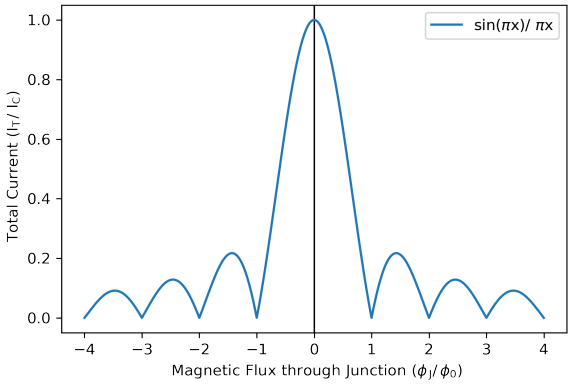
B. Superconducting Quantum Interference Devices

Superconducting Quantum Interference Devices, usually abbreviated as SQUID, have become instrumental in Condensed Matter Physics ever since their invention in 1964 [13], thanks to their high sensitivity for the measurement of low-frequency magnetic fields. Apart from this field in which they were initially discovered, one of the main focuses at the moment is the application of SQUID arrays in biomagnetism, with particular emphasis on neuromagnetism and magnetocardiography [14].

The device is formed by interrupting a superconducting ring, i.e. a material which supports resistanceless electric transport, by one or two Josephson junctions. The latter are junctions between two superconductors separated by a thin insulating barrier, which allows Cooper pairs to tunnel from one superconductor to the other, even with no applied voltage [15]. The fact that electron pairs could



(a) Screening current



(b) Fraunhofer-like interference of current

FIG. 4: (a) Shows that the screening current is periodic in the applied flux, with a period equal to one flux quantum. (b) Variation of the total current through the SQUID as a function of the magnetic flux applied, when the width of the Josephson junction is not neglected.

tunnel through such a barrier only due to the phase difference of their wavefunctions across it was suggested by Josephson in 1962 [16] and observed one year later by Anderson and Rowell [17].

Schematically, such a device can be seen in FIG. 3, where the Josephson tunnel junctions are shown in red. The simplest experiment to perform with this circuit is to pass a bias current I_B over the ring and to measure the voltage V that appears across it: if the SQUID is symmetrical with identical junctions and no magnetic field is applied, then the current divides evenly and half of it passes through each junction before recombining on the other side. As long as the total current flowing through the SQUID does not exceed the critical current of the Josephson junctions (i.e. the maximum zero–resistance current that can pass through them), a supercurrent flows through the device.

However, whenever a magnetic field is applied inside the ring, the superconducting loop tries to oppose the applied field by generating a screening current I_S, which flows around the loop effectively canceling out the net

flux in the ring. Considering that the bias current splits in $I_B = I_1 + I_2$, this means that in one of the branches I_S flows in the same direction as I_1, giving rise to a total current $I_1 + I_S$, while in the other branch I_S flows in the opposite direction to I_2, which gives $I_2 - I_S$. When this new current in either of the branches exceeds the critical current of the junction, the SQUID becomes resistive and therefore a nonzero voltage can be measured across it.

Since the magnetic flux enclosed by the superconducting loop must be an integer number of flux quantum Φ_0, when the external field is increased above $\Phi_0/2$, the screening current changes sign as it is more energetically favorable for the SQUID to increase the total flux through the loop to Φ_0 instead of trying to cancel the applied one. Thus, the screening current changes direction periodically, every time the flux increases by a half-integer multiple of Φ_0, as can be seen in FIG. 4(a).

The screening current also creates an additional Josephson phase in the two branches, ϕ_1 and ϕ_2:

$$\phi_1 \propto \frac{2e}{\hbar}\int_{\text{path}_1} \boldsymbol{A}\cdot d\boldsymbol{l} \tag{3}$$

$$\phi_2 \propto \frac{2e}{\hbar}\int_{\text{path}_2} \boldsymbol{A}\cdot d\boldsymbol{l} \tag{4}$$

$$\phi_1 - \phi_2 \propto \frac{2e}{\hbar}\oint_{\text{loop}} \boldsymbol{A}\cdot d\boldsymbol{l} = \frac{2\pi\Phi_{\text{loop}}}{\Phi_0} \tag{5}$$

where $\boldsymbol{A}$ is the vector potential and Φ_{loop} is the magnetic flux through the loop. Note that the above relations are a straightforward result from Schrödinger's equation in a magnetic field.

The supercurrent flowing through each junction is related to this Josephson phase by the current-phase relation $I = I_C \sin\phi$ [18], such that the total current leaving the SQUID is given by:

$$I_T = I_1 + I_2 = I_C\sin\phi_1 + I_C\sin\phi_2 = 2I_C\cos\left(\frac{\phi_1-\phi_2}{2}\right)\sin\left(\frac{\phi_1+\phi_2}{2}\right) \rightarrow I_T^{\max} = 2I_C\left|\cos\left(\frac{\pi\Phi_{\text{loop}}}{\Phi_0}\right)\right|. \tag{6}$$

Here we used of the result obtained in Equation (5) and of the fact that $\sin(x)$ takes a maximum value of 1.

However, the phase across the junction is no longer homogeneous when considering that it has a finite thickness t and width w, since the magnetic field penetrating through the junction creates a phase gradient across its width [19]. In this case, the critical current takes a similar form to the well-known Fraunhofer interference for a narrow rectangular slit:

$$I_T^{\max} = I_C\left|\frac{\sin(\pi\Phi_J/\Phi_0)}{\pi\Phi_J/\Phi_0}\right|, \tag{7}$$

where $\Phi_J = B\cdot w\cdot t$ is the flux through the junction. This phenomenon is observed when taking measurements over a larger range of magnetic fields and it is schematically shown in FIG. 4(b).

II. Electrical Measurements

A. Low Temperature Measurements of Graphene

The measurements of the device were done using a four-terminal measurement set-up (see FIG. 2). This set-up allows for the measurement of the longitudinal voltage along the device as well as the transverse voltage which will be the Hall voltage.

The device itself was composed of a layer of graphene with multiple contacts attached at various points. Two current carrying contacts were connected at each end of the graphene, as well as seven other contacts. Optical microscopy images of different magnifications can be seen in FIG. 5(a), 5(b) and 5(c), where contacts were labelled alphabetically, A to N.

Terminal	Contact
1	G
2	I
3	J
7	K
8	L
9	M
10	A
11	B
24	F

(a) Hall bar contacts

Terminal	Contact
17	C/D
18	C/D
20	N
21	E
23	E
15	back gate

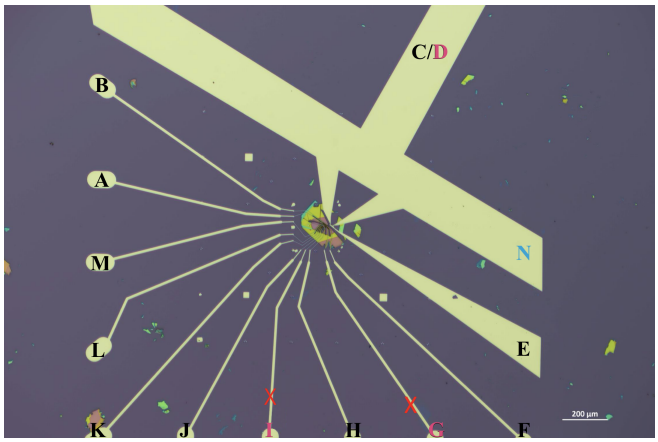
(b) SQUID contacts

TABLE I: (a) Graphene Hall bar and (b) SQUID terminals and their corresponding channels.

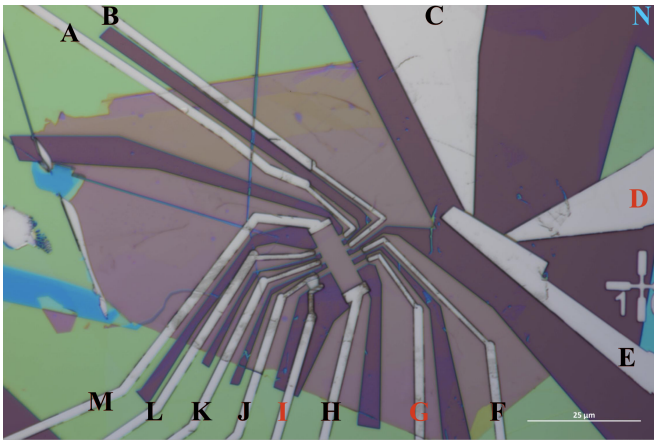
Before the device was loaded into the fridge to be properly cooled, the connections had to be tested to ensure they were intact. The device was temporarily loaded into the fridge and external terminals were assigned to each contact. The exact assignments for each contact are shown in the Table I.

Channel "I" was known to have shorted to channel "J" before the test began. Terminal 2 was selected to be the ground while terminal 9 was used as the current source. Terminals 7 and 11 were chosen for the points at which the transverse measurements would be taken, while 10 and 24 were selected for longitudinal measurements. 1 and 10 would have been better choices for the longitudinal measurements as the potential difference would be greater owing to the larger distance between measurement sites. However, the associated contacts were noted to have been faulty before the test had begun.

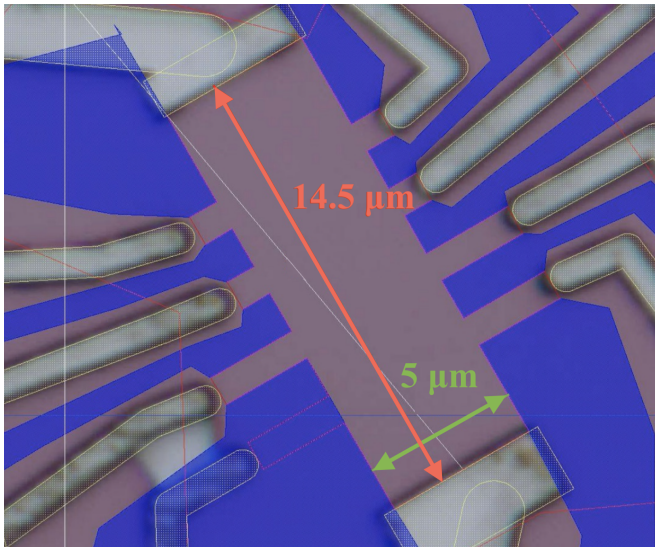
Following this, the device was loaded into the fridge



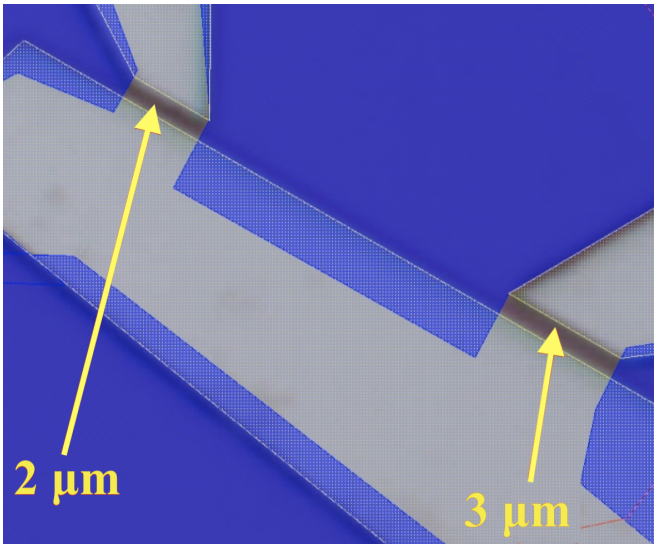
(a) Optical image of devices - magnification x50



(b) Optical image of devices - magnification x100



(c) Hall Bar - magnification x200



(d) SQUID - magnification x200

FIG. 5: Optical images of the two devices and their channels, i.e. the graphene Hall bar and the SQUID, taken with various microscope magnifications. Note that the length scales are (a) 200µm and (b) 25µm.

and slowly cooled overnight until it reached a temperature of ≈ 0.02 K. Measurements were taken using the BadgerLoop program with the inputs from lock-ins being "VT.DM1-1.R" and "VT.DM1-2.Phase" for the transverse voltage, "VL.DM1-1.R" and "VL.DM1-2.Phase" for the longitudinal one, and the notable outputs being the back gate voltage and the applied magnetic field within the fridge. The measurements were split across two readings. More steps in voltage were taken at higher fields, which was done to examine the more interesting phenomena at higher field strengths. The step count was decreased once the field was at 4 T, as the visible Landau levels were less pronounced at lower magnetic fields. Once measurements were completed, they were exported as text files and imported into a Jupyter notebook.

The code was used to analyse the data and to calculate the Hall resistance from the lock-in amplitude, which was measured in units of V_{rms}, and the AC current used. From this, the resistance and then conductance could be computed and thus used to check whether the QHE was occurring. The former was obtained using Equation (8), and the latter, measured in units of conductance quantum e^2/h, in (9):

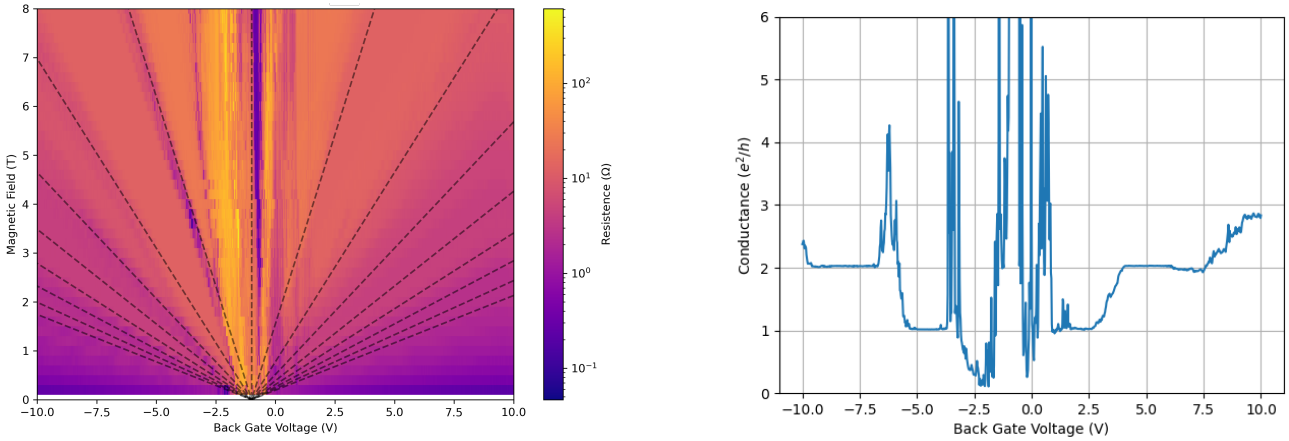
$$R_{AC} = \frac{LI.R}{I_{AC}} \tag{8}$$

$$G_{AC} = \frac{1/R_{AC}}{e^2/h}. \tag{9}$$

The values for resistance were used to generate a colourmap by plotting magnetic field on the y-axis and back gate voltage on the x-axis. For each magnetic field and voltage we calculated the corresponding resistance and plotted it by associating colours to these values.

B. Low Temperature Measurements of a SQUID

Our second device was a SQUID triangular in shape, consisting of two superconductor-graphene-superconductor tunnelling junctions, where graphene was covered by two NbTi electrodes with a narrow gap of ≈ 0.5µm between them.



(a) Landau fan

(b) Conductance steps

FIG. 6: (a) Resistance colourmap with respect to applied magnetic field and back gate voltage, where the black dashed lines separate regions of different filling factors. (b) Plot of conductance against gate voltage for $B = 8$T, which clearly shows the quantisation of the Hall conductivity in units of e^2/h.

The measurement of this device was done using a similar set-up to the one sketched in FIG. 3: the two (asymmetrical) Josephson tunnelling junctions are shown in the optical image from FIG. 5(d), and the traingular SQUID loop can be observed in FIG. 5(a).

It should be noted that the faulty contacts are clearly labelled in red in FIG. 5(a) and 5(b), unlike the operational ones which are shown in black. Although channels "C" and "D" seem distinct in FIG. 5(b) at a x100 magnification, they are in fact connected to the same part of the SQUID, as can be clearly seen under a lower magnification in FIG. 5(a). Channel "N" appears in cyan and it is not ordered alphabetically because it was added later in the experiment to enable 4-terminal measurements for the SQUID, when channel "D" was tested again and proved to be faulty.

Terminals 17 and 23 were selected for measuring the current, while terminals 18/20 and 21 were used as the voltage source. According to Table I, channel "E" corresponds to both terminals 21 and 23, and it has introduced a resistance in series with the SQUID as a result of being used for both current and voltage measurements simultaneously.

Once the sample reached a temperature of ≈ 0.02K in the fridge overnight, measurements were taken using again the BadgerLoop program. The main input from the lock-in amplifier was the voltage across the SQUID "SQUID.DM1-1.R", which give a resistence:

$$R_{\text{SQUID}}(\Omega) = \frac{\text{SQUID.DM1-1.R}}{\text{SQUID.Osc1.A} \cdot 10^{-9} A/V}. \tag{10}$$

The outputs needed for plotting the data were the biasing voltage "SQUID.Aux1Offset", which is converted from V_{rms} into biasing current measured in µA using Equation (11), and the current through the solenoid "Magnet.IOut" which allowed us to determine the magnetic field via Equation (12).

$$I_{\text{SQUID}}(\mu\text{A}) = \text{SQUID.Aux1Offset} \cdot 10^{-6} A/V \tag{11}$$

$$B(\text{T}) = \frac{\text{Magnet.IOut}}{13.72\, A/T} \tag{12}$$

These three parameters were used to generate a colourmap by plotting magnetic field on the x-axis and the bias current (i.e. I_B in FIG. 3) on the y-axis, with the corresponding resistance at each point as the argument of the colourmap.

III. Results and Discussion

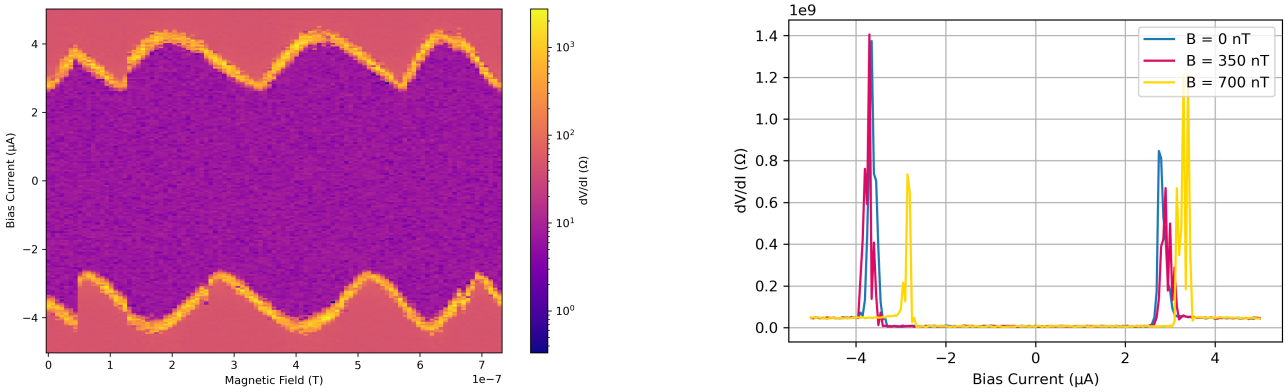
A. Graphene Hall Bar

As can be seen in FIG. 6(a), the device tested in this investigation clearly exhibits the QHE as evidenced by the distinct Landau levels in the colourmap, as well as the presence of the plateaus of conductance in FIG. 6(b).

The shape visible in FIG. 6(a) is known as a "Landau Fan". Within the plot, there are visible edges of decreasing resistance, which correspond to the plateaus seen in FIG. 6(b) where there is a sudden increase in conductance with increasing back gate voltage, i.e. as a Landau level is crossed. These edges are fit with a set of black dashed lines, which have a gradient corresponding to the filling factor of each Landau level visible. If the QHE is occuring as expected, one would expect these lines to have an incrementally decreasing gradient proportional to some constant a. The decrease would be proportional to an ever increasing integer N, and therefore the gradient of these lines could be modelled as

$$\text{Gradient} = a/N, \tag{13}$$

where N is the filling factor of each Landau level. Note that this equation is also proportional to the degeneracy of these Landau levels (e.g. $\pm a/2$, $\pm a/4$, $\pm a/6$ etc. for spin degeneracy of 2), but FIG. 6(b) shows that the degeneracy in our case is 1.



(a) Colourmap of differential resistance

(b) Differential resistance plots

FIG. 7: (a) Colourmap of differential resistance as a function of both bias current and magnetic field. (a) Differential resistance as a function of the bias current at small magnetic fields $B = 0\,\mathrm{nT}$, $B = 350\,\mathrm{nT}$ and $B = 700\,\mathrm{nT}$.

During analysis of the results, it was deduced that the value of the constant a was ≈ 1.6 (measured in units T/V). This was found by continually changing the value of a until the straight lines were as close to the edges of the Landau levels as possible. It should be noted there is some deviance between the lines on the left hand side of the fan which is likely due to the Landau levels not being exactly evenly spaced in the plot. Another reason could be that because the exact value of the charge neutrality point is not known, the lines could be off along the x-axis.

As the first Landau level is slightly off-centre in the plot, sitting at approximately $-1V$, it is reasonable to conclude that the graphene in the sample device is likely to be p-doped. As the gate voltage in a transistor determines the number of each type of charger carrier within the device, an offset such as this would imply a higher number of hole charge carriers within the sample device.

Knowing the gradient of the lines separating the distinct regions of the Landau fan also allows us to calculate the carrier density in graphene:

$$n = \frac{Q}{eA} = \frac{CV_G}{eA} = \frac{C}{A}\cdot\frac{1}{e}V_G \tag{14}$$

$$B = \underbrace{\left(\frac{\phi_0}{N}\cdot\frac{1}{e}\cdot\frac{C}{A}\right)}_{=a/N \text{ from FIG. 6(a)}} V_G \tag{15}$$

$$n = a\,\frac{\cancel{e}}{\phi_0}\cdot\frac{1}{\cancel{e}}V_G = \frac{a}{\phi_0}V_G \tag{16}$$

where C/A is the capacitance per unit area of the graphene device. Knowing the relation between carrier density and back gate voltage, we could rescale the x-axis in both plots in FIG. 6 by $1.6/\phi_0$ and therefore reproduce the same results in more physically sensible units.

The graphene used in the device is more likely to be a multilayer sample than a monolayer or bilayer as the increase in conductance differs from what one would typically expect from such sample thicknesses. The value of conductance for a monolayer would increase in larger steps owing to the 4-fold degeneracy present and bilayers would also have larger step sizes with their degeneracy of 2. Many layer samples however, would only have a degeneracy of 1 which is what can be seen in FIG. 6(b). Additionally, in FIG. 6(a) there is a notable distortion in the centre of the plot which is a noted feature of Landau fans derived from many layer samples [20].

B. SQUID

In order to test the functionality of the SQUID, we started by plotting the differential resistance dV/dI as a function of the bias current at different magnetic fields, as shown in FIG. 7(b). From these plots we can infer that the critical current is $\approx 3\mu$A and that our device is indeed superconducting.

The colourmap given in FIG. 7(a) extend this analysis for multiple values of magnetic field, as it shows how the SQUID differential resistance varies with both bias current and magnetic field. It should be noted that the three plots in FIG. 7(b) are nothing more than vertical cuts at different magnetic fields in the colourmap. As discussed earlier, the oscillatory behaviour of the SQUID current is explained by the periodic change in sign of the screening current shown in FIG. 4(a). Since the period of oscillation of the screening current is ϕ_0, in terms of the magnetic field that should be ϕ_0/A_{loop}. Knowing the period ΔB allows us to determine the area of the loop:

$$A_{\text{loop}} = \frac{\phi_0}{\Delta B} \approx \frac{2.067\cdot 10^{-15}\mathrm{Tm}^2}{2.5\cdot 10^{-7}\mathrm{T}} = 8,271\,\mu\mathrm{m}^2. \tag{17}$$

This area is of the same order of magnitude as the area of the triangle we obtain geometrically from FIG. 5(a): $A_\Delta = b\cdot h/2 \approx 150\mu\mathrm{m}\cdot 200\mu\mathrm{m}/2 = 15,000\mu\mathrm{m}^2$.

Increasing now the range of magnetic field, we can observe the Fraunhofer-like interference pattern due to the finite size of the tunnelling junctions, as shown FIG. 8. Similar to before, the period of the enveloping oscillations at larger magnetic fields ΔB_J allows us to estimate the area of the Josephson tunneling junctions, by making use of the expression previously given in Equation (7):

$$A_J = \frac{\phi_0}{\Delta B_J} \approx \frac{2.067\cdot 10^{-15}\mathrm{Tm}^2}{0.5\cdot 10^{-3}\mathrm{T}} = 4.134\,\mu\mathrm{m}^2. \tag{18}$$

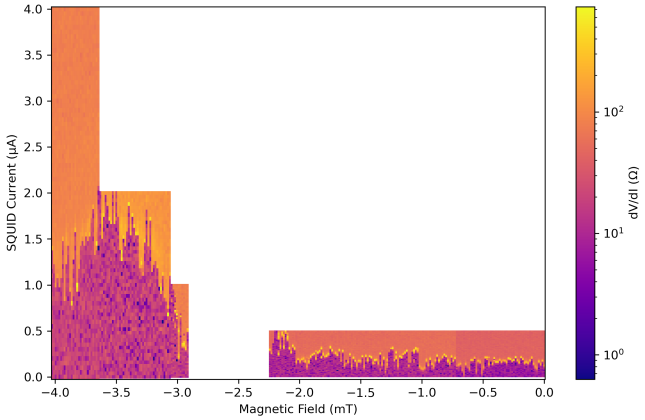


FIG. 8: Colourmap of differential resistance composed of multiple plots patched together to illustrate the Fraunhofer-like interference pattern of the critical current under varying magnetic field

Although looking at FIG. 5(d) we might expect a slightly smaller area, i.e. $A_{\square} \approx 0.5\mu\text{m} \cdot 2\mu\text{m} = 1\mu\text{m}^2$ or $A_{\square} \approx 0.5\mu\text{m} \cdot 3\mu\text{m} = 1.5\mu\text{m}^2$, our result is still of the same order of magnitude.

It should be noted that instead of being centered around zero magnetic field like FIG. 4(b), our plot has its peak at $\approx -3.5\,\text{mT}$. This means there is magnetic flux trapped in the magnet of our apparatus, which effectively shifts all our measurements by $B = -3.5\,\text{mT}$ from the theoretical predictions. The sharp change in the colourmap from FIG. 7(a) around $B = 0.5 \cdot 10^{-7}\,\text{T}$ is also an artefact of this.

IV. Conclusions

The results of the investigation prove the occurrence of the QHE in graphene. More specifically, the plateaus in conductance shown in FIG. 6(b) change in integer steps in units of e^2/h. This is clear evidence of the integer QHE which is known to occur in graphene. The charge neutrality point was found to be at $\approx -1\text{V}$ implying the graphene in the device was a p-type semiconductor. There is also evidence that the graphene in the sample is a multilayer, owing to the shape of the landau fan plot and the step size in conductance.

Our measurements taken on the SQUID not only confirmed its superconducting behaviour at low temperatures by observing the zero differential resistance at a bias current lower than the critical one, but also replicated the oscillatory behaviour of the bias current as a function of magnetic field, as predicted by the theory. These measurements also allowed us to estimate the area of the SQUID loop and its two tunnelling junctions. These calculations proved to be consistent with our prior estimations based on the magnified images of the sample, and therefore confirmed the functionality of the device.

A possible improvement to the investigation would be to run the measurements over more sweeps. While this is unlikely to affect the shape of the Landau fan or of the SQUID oscillations, it could provide a more clear image.

Acknowledgments

Thank you so much (Jon and Mike) for assigning us this project and for all the time and effort you invested into it! Also thank you George for taking the time to demonstrate the second sound experiment to us.

[1] E. H. Hall *et al.*, On a new action of the magnet on electric currents, American Journal of Mathematics **2**, 287 (1879).
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